A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology

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Abstract

1 Introduction

Full House Modeling provides a novel foundation for comparative analysis of individuals evaluated across differing contexts. This foundation originated in an analysis comparing baseball players across eras [Yan et al., 2025a]. The historical significance of this analysis was further studied and presented in Burgos Jr et al. [2025]. In baseball these analyses are game changing and gaining traction in the broader baseball analytics community. Full House Modeling was expanded upon in an investigation of the impact of historical figures [Cohen et al., 2025]. **needs to be edited and made much more interesting**

In this article we formalize Full House Methodology. Under this formalization we study probabilities of observing paradoxes between latent aptitude and observed achievement in comparative analyses between individuals being compared across contexts. Here, the paradox is that an individual from one context can have higher aptitude and simultaneous lower observed achievement than an individual from another context. We provide explicit probabilities of these paradoxes in two settings: 1) when observed achievement is independent and identically distributed (iid); 2) when the observed statistics across contexts follow different normal distributions but are still independent. We also study bounds to the paradoxes in more general settings. We also demonstrate that when observed achievement is normally distributed, the paradox probabilities can be reliably estimated via plug-in.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent $\mathbb{R}$-valued aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{iid}{\sim} F_X$. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes

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not including the component j. We then define an indicator function $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right)$ as a determined and known system inclusion function for population i where a 1 indicates inclusion and a 0 indicates exclusion. We now suppose that an individual j included in system i has an observed $\mathbb{R}$-valued statistic $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$, $n_i \leq N_i$. As we will see later, the observed $Y_{i,j}$ values are one-to-one realizations of $X_{i,j}$, provided that $g_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1$. Therefore $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$.

Let $X_{i,(j)}$ and $Y_{i,(j)}$ denote the jth order statistic for system i. Here, larger numbers will correspond to a notion of "better." Thus $X_{i,(j)} > X_{i,(k)}$ whenever $j > k$, for example. We now form (Y, X) pairs of random variables for each individual included in the system. These pairs are of the form $(Y_{i,(j)}, X_{i,(r_{i,j})})$ where $r_{i,j}$ indexes the aptitude of the jth highest value observed from the system. As an example, let $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$, then the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs take the form $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ for $j = 1, \ldots, n_i$.

The key Full House Modeling is the explicit connection between the elements forming the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs. Full House Modeling states this connection as

$$\begin{aligned} Y_{i,(j)} &= F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}}\left(X_{i,(r_{i,j})}\right), \\ X_{i,(r_{i,j})} &= F^{-1}_{X_{i,(r_{i,j})}} \circ F_{Y_{i,(j)}}\left(Y_{i,(j)}\right), \end{aligned} \tag{1}$$

where $1_i(X_{i,(r_{i,j})}, \mathbf{X}_{i,-(r_{i,j})}) = 1$, and both F_{Y_i} and F_X are continuous and invertible. We summarise Full House Modeling in the definition below.

Definition 2.1 (Full House Model)**.** *Consider populations with sizes N_i and system sizes n_i, $N_i \geq n_i$. Let a system inclusion function $1_i(\cdot, \cdot)$ be specified. Let $X_{i,j} \overset{iid}{\sim} F_X$, $j = 1, \ldots, N_i$. Let $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$. Form pairs $(Y_{i,(j)}, X_{i,(r_{i,j})})$, $j = 1, \ldots, n_i$ for all individuals such that $1_i(X_{i,(r_{i,j})}, \boldsymbol{X}_{i,-(r_{i,j})}) = 1$. Then the Full House Model generates $Y_{i,(j)}$ values following (1) where F_{Y_i} and F_X are continuous and invertible.*

Note that Definition 2.1 fixes F_X but allows F_{Y_i} to vary across systems.

Remarks:

1. The definition of a Full House Model immediately implies structural missingness. Every $X_{i,j}$ such that $1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ will not have a corresponding Y value. In the following sections we will consider probabilities involving both X and Y quantities. To handle this structural missingness, we will suppose that $Y_{i,j}$ values that are not observed are not P-measurable.

3 Paradoxes Under Identical Distributions

In this section we will provide explicit paradox probability results in a Full House Modeling setting involving two populations with the additional specification that $F_{Y_i} = F_Y$, $i = 1, 2$ in Definition 2.1. The following definition formally defines the notion of paradox used throughout this article.

Definition 3.1. *Consider the Full House Model in Definition 2.1 with two populations denoted by $i = 1, 2$. We say a paradox has occurred when both $X_{1,(r_{1,j})} > X_{2,(r_{2,k})}$ and $Y_{1,(j)} < Y_{2,(k)}$ have occurred. The paradox probability is defined as*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right). \tag{2}$$

As a simple illustrative example of the above, let $r_{i,t} = N_i - n_i + t$ and set $j = n_1$ and $k = n_2$, and further suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then the paradox probability (2) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ when $N_1 \to \infty$. We now introduce a convenient representation which allows the paradox probability to be expressed in terms of the latent aptitudes alone.

Lemma 3.2. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Then,*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right), \tag{3}$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$.

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{1,2,j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. □

Our first formal result provides conditions that allow for an explicit paradox probability result. Those conditions require the following definition.

Definition 3.3. *Let $\alpha, \beta > 0$. Then the regularized incomplete beta function is defined as*

$$I_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} \int_0^x u^{\alpha-1}(1-u)^{\beta-1} du,$$

where $B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta coefficient.

Note that the regularized incomplete beta function is the cumulative distribution function of a Beta(α, β) distribution.

Theorem 3.4. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Denote by I the regularized incomplete beta function. Further assume that*

$$I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \geq 0, \qquad \text{for all } t \in (0,1). \tag{4}$$

Under these assumptions,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&= \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2-n_1-p+k+1)}{B(k, n_2-k+1)} - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2-N_1-p+r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

Remarks:

1. The result of Theorem 3.4 is distribution free. It only depends on the sample sizes n_i, N_i, $i = 1, 2$, the system inclusion function, and the indexes of the order statistics under comparison. The distribution on Y is eliminated by Lemma 3.2. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = I_{j,n_1-j+1} \circ F_Y(y)$. With this in mind, from Lemma 3.2 we have that

$$\begin{aligned} h_{1,2,j,k}(x) &= F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(x) \\ &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ \left(F_Y \circ F_Y^{-1}\right) \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x) \\ &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x). \end{aligned}$$

   The above is the upper-bound of an integral with respect to Xs in the proof of Theorem 3.4. The influence of the distributional specification on Y does not influence the calculation of the paradox probability result in Theorem 3.4.

2. The distribution on X is eliminated when calculating the paradox probability (2) in its Lemma 3.2 representation, $\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$. These calculations involve double integrals with respect to densities for $F_{X_{1,(r_{1,j})}}$ and $F_{X_{2,(r_{2,k})}}$. One integral evaluation and a reparameterization remove the specification of F_X from the probability calculation as seen in the proof of Theorem 3.4.

3. The steps mentioned in the last sentence of the previous remark allow the probability calculation for $\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$ to simplify to

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.$$

   The technical condition (4) ensures that this integral defines a non-negative probability in strong terms. We demonstrate with analytic and numeric arguments when (4) is satisfied and the results of Theorem 3.4 therefore hold.

Proof. First, Lemma 3.2 allows us to write

$$\text{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{1,2,j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. o see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = g_{n,j} \circ F_Y(y)$, where we write $g_{n,j}(x) = I_{j,n-j+1}(x)$ for ease of notation. This expression of distribution functions allows us to write

$$\begin{aligned} &\text{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right) \\ &= \int_{-\infty}^{+\infty} \left\{ \int_x^{F_X^{-1} \circ g^{-1}_{N_1,r_{1,j}} \circ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \end{aligned}$$

$$
\begin{aligned}
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1}\circ g_{N_1,r_{1,j}}^{-1}\circ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x)} d\ g_{N_1,r_{1,j}}(F_X(y)) \right\}\ f_{X_{2,(r_{2,k})}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x) - g_{N_1,r_{1,j}}\circ F_X(x) \right\}^+\ f_{X_{2,(r_{2,k})}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}\circ g_{N_2,r_{2,k}}\circ F_X(x) - g_{N_1,r_{1,j}}\circ F_X(x) \right\}^+\ d\ g_{N_2,r_{2,k}}(F_X(x)) \\
&= \int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^+\ dt.
\end{aligned}
$$

Condition (4) states that

$$
\int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^+\ dt = \int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}\ dt.
$$

Thus,

$$
\begin{aligned}
&\int_0^1 \left\{ g_{n_1,j}\circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}}\circ g_{N_2,r_{2,k}}^{-1}(t) \right\}^+\ dt \\
&\quad = \int_0^1 g_{n_1,j}(x)\ dg_{n_2,k}(x) - \int_0^1 g_{N_1,r_{1,j}}(x)\ dg_{N_2,r_{2,k}}(x) \\
&\quad = \int_0^1 I_x(j, n_1-j+1)\frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\ dx \\
&\qquad - \int_0^1 I_x(r_{1,j}, N_1-r_{1,j}+1)\frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(k,N_2-k+1)}\ dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p(1-x)^{n_1-p}\frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\ dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p(1-x)^{N_1-p}\frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k},N_2-r_{2,k}+1)}\ dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2+n_1-p-k+1)}{B(j,n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k},N_2-r_{2,k}+1)}.
\end{aligned}
$$

□

Often it is of interest to compare the individuals with the highest aptitudes in their respective populations. Below we state the paradox probability in such setting using a Full House Model with two populations with the specification that $F_{Y_1} = F_Y$ for $i = 1, 2$.

Corollary 3.4.1. *Suppose that the conditions of Theorem 3.4 hold. Suppose that* $1_1(X_{1,(N_1)}, \boldsymbol{X}_{1,-(N_1)}) = 1_2(X_{2,(N_2)}, \boldsymbol{X}_{2,-(N_2)}) = 1$. *Then,*

$$
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \le \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.
$$

A similar result holds for the individuals with the lowest latent aptitudes in the population.

Corollary 3.4.2. *Suppose that the conditions of Theorem 3.4 hold. Suppose that* $1_1(X_{1,(1)}, \boldsymbol{X}_{1,-(1)}) = 1_2(X_{2,(1)}, \boldsymbol{X}_{2,-(1)}) = 1$. *Then,*

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

We now demonstrate a setting in which technical condition (4) holds. This setting is based on a configuration of n_j, N_j, $j = 1, 2$ under the assumption that the systems are populated with individuals from the source populations who have the highest aptitudes.

Proposition 3.5. *Consider the Full House Model in Definition 2.1 with two populations, where* $F_{Y_i} = F_Y$ *for* $i = 1, 2$, *and let the system inclusion function be* $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$. *Denote by* I *the regularized incomplete beta function. Suppose that* $\frac{n_1}{n_2} \leq 1$ *and* $\frac{N_2}{N_1} \leq 1$, *and set* $k = j$. *Under these assumptions, we conclude that technical condition (4) holds and the conclusion of Theorem 3.4 applies.*

Proof. Specification of $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$ implies that $r_{i,j} = N_i + 1 - j$.

When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0, 1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased [Askitis, 2021], or n_2 or N_1 are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned} & I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t. \end{aligned}$$

Technical condition (4) holds and the conclusion of Theorem 3.4 applies. □

4 Paradox Under Different Distribution

4.1 General bounds

Definition 4.1. *We will say that* Y_2 *is first order* ε*-weakly stochastically dominant over* Y_1 *if, for* $\varepsilon > 0$, $\mathrm{P}(Y_2 \leq y) \leq \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ *for all* y *and* $\mathrm{P}(Y_2 \leq y) < \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ *for some* y.

Proposition 4.2. *Suppose that the system inclusion mechanism has the property that* $g(X_{1,(r_{1,k})}) = 1$ *and* $g(X_{2,(r_{2,j})}) = 1$, *and that the Full House Model is of the form*

$$Y_{i,j} \sim F_{Y_i}(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i,$$

$$X_{i,j} \sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i,$$

where $F_{Y_i}^{-1}(y)$ and $F_X^{-1}(x)$ exist. Further suppose that Y_2 is first order ε-weakly stochastically dominant over Y_1 for some $\varepsilon > 0$. Then,

$$\begin{aligned} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2} &\leq \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\leq n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2} \end{aligned} \tag{5}$$

Proof. First, observe that the assumption of first order ε-weakly stochastically dominant gives

$$\left(F_{Y_{2,(n_2)}}(y)\right)^{n_1/n_2} \leq \left(F_{Y_1}(y)\right)^{n_1} \leq \left(F_{Y_2}(y)+\varepsilon\right)^{n_1} = \left(\left(F_{Y_{2,(n_2)}}(y)\right)^{1/n_2} + \varepsilon\right)^{n_1},$$

and this implies that

$$F_X^{-1}\left(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}}(X_{2,(N_2)})\right) \leq h_{1,2}(X_{2,(N_2)}) \leq F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(X_{2,(N_2)})\right)^{1/n_2} + \varepsilon\right)^{n_1}\right).$$

In combination with Corollary 3.4.1 and the proof of Corollary 3.4.1, the first inequality yields

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \geq \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}.$$

This verifies the first inequality of (5). We now verify the second inequality of (5). Now,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad\leq \int_{-\infty}^{\infty}\left[\int_x^{F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(x)\right)^{1/n_2}+\varepsilon\right)^{n_1}\right)} f_{X_{1,(N_1)}}(y)dy\right] f_{X_{2,(N_2)}}(x)dx \\ &\quad= \int_{-\infty}^{\infty}\left[\left(\left(F_X(x)\right)^{N_2/n_2}+\varepsilon\right)^{n_1} - F_X(x)^{N_1}\right] N_2 F_X(x)^{N_2-1} f_X(x)dx \\ &\quad= \int_{-\infty}^{\infty}\left(\left(F_X(x)\right)^{N_2/n_2}+\varepsilon\right)^{n_1} N_2 F_X(x)^{N_2-1} f_X(x)dx - \frac{N_2}{N_1+N_2} \\ &\quad= \int_0^1 \left(t^{1/n_2}+\varepsilon\right)^{n_1} dt - \frac{N_2}{N_1+N_2} \\ &\quad= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2} \end{aligned}$$

□

4.2 Normal distributions

In this example we demonstrate a case where explicit paradox probabilities (2) when a Full House Model in Definition 2.1 additionally specifies that $Y_i \sim N(\mu_i, \sigma_i^2)$, and suppose that the system inclusion mechanism has the property that $1_i(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$ for $1 = 1, 2$. Let $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ and covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}, \tag{6}$$

defined with the following structure

$$\begin{aligned} \Sigma_{11} &= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1 \times n_1} \qquad \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1)\times n_1} \\ \Sigma_{22} &= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} . \end{aligned} \tag{7}$$

Let Φ be the cumulative distribution function of the standard normal distribution. Our paradox probability result is presented in the Thereom below.

Theorem 4.3. *Assume a Full House Model in Definition 2.1 and additionally specify that* $Y_i \sim N(\mu_i, \sigma_i^2)$, *and suppose that the system inclusion mechanism has the property that* $1_i(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}) = 1$ *for* $1 = 1, 2$. *Let* $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ *where* $\mathbf{\Sigma}$ *is as defined in* (6) *and* (7). *Assume that* $\Phi(\frac{\Phi^{-1}(t^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1})^{\frac{n_1}{N_1}} > t$ *for all* $t \in (0,1)$. *Then,*

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad = n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \dots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \dots, 0)) - \frac{N_2}{N_1+N_2}, \end{aligned} \tag{8}$$

where $F_{\mathbf{W}}$ *is the distribution function of* $\mathbf{W}$, *and the first* n_1 *elements of* $\mathbf{w}$ *are* $\frac{\mu_2-\mu_1}{\sigma_1}$ *and the last* $n_2 - 1$ *elements are 0.*

Proof. First let

$$\begin{aligned} h_{1,2}(x) &= (g_1 \circ f_2)(x) \\ &= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\ &= F_X^{-1}((\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{\frac{n_1}{N_1}})(x). \end{aligned}$$

With this definition of $h_{1,2}(x)$ we can write

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right)$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} N_1 (F_X(y))^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{F_X^{-1}(\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1})^{\frac{n_1}{N_1}})(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{-\infty}^{+\infty} ((\Phi(\frac{(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_0^1 (\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1} - t^{\frac{N_1}{N_2}})\ dt$$

$$= \int_0^1 (\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}\ dt - \frac{N_2}{N_1+N_2}$$

$$= \int_{-\infty}^{+\infty} \Phi(\frac{\sigma_2 x+\mu_2-\mu_1}{\sigma_1})^{n_1} (n_2 \Phi(x)^{n_2-1}\phi(x))\ dx - \frac{N_2}{N_1+N_2}$$

$$= n_2 \int_{-\infty}^{+\infty} \mathbb{P}(Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \le -\frac{\mu_2-\mu_1}{\sigma_1},$$

$$Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 | X = x)\phi(x)\ dx - \frac{N_2}{N_1+N_2}$$

$$= n_2 \mathbb{P}(Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \le -\frac{\mu_2-\mu_1}{\sigma_1}, Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0)$$

$$- \frac{N_2}{N_1+N_2}$$

$$= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1+N_2}.$$

The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (6) and (7). The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last n_2-1 elements are 0. □

We now show that plug-in estimates for normal distribution parameters μ_i, σ_i, $i = 1, 2$ allow for reliable estimate of the paradox probability (8). Our approach requires properties for matrices of the form

$$\Sigma = \begin{bmatrix} c+1 & c & \ldots & c \\ c & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c \\ c & \ldots & c & c+1 \end{bmatrix}_{n \times n}.$$

These properties are

$$\det(\Sigma) = cn + 1, \tag{9}$$

and

$$\Sigma^{-1} = \frac{1}{c \cdot n + 1} \begin{bmatrix} c \cdot (n-1) + 1 & -c & \dots & -c \\ -c & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -c \\ -c & \dots & -c & c \cdot (n-1) + 1 \end{bmatrix}_{n \times n}, \tag{10}$$

where (10) follows from an application of the Woodbury identity [Golub and Van Loan, 2013].

Before we state our result, we introduce the Lemma 4.4

Lemma 4.4. *Let* $\mathbf{W} \sim \mathcal{N}_{n_1+n_2-1}(\mathbf{0}, \mathbf{\Sigma})$ *be a multivariate normal random vector with mean vector* $\mathbf{0}$ *and covariance matrix* $\mathbf{\Sigma}$ *as defined in* (6) *and* (7). *We denote the cumulative distribution function (CDF) of* $\mathbf{W}$ *by* $\mathbf{\Phi}(\mathbf{w}; \mathbf{\Sigma})$, *where* $\mathbf{w} \in \mathbb{R}^n$ *is the quantile vector at which the CDF is evaluated.*

Define

$$\begin{aligned} c = \max \Big\{ & \Phi(W_{n_1+n_2-1} = w_{n_1+n_2-1} | W_1 = w_1, W_2 = w_2, \dots, W_{n_1+n_2-2} = w_{n_1+n_2-2}), \\ & \Phi(W_{n_1+n_2-2} = w_{n_1+n_2-2} | W_1 = w_1, W_2 = w_2, \dots, W_{n_1+n_2-3} = w_{n_1+n_2-3}), \dots, \\ & \Phi(W_2 = w_2 | W_1 = w_1), \Phi(W_1 = w_1) \Big\}, \end{aligned} \tag{11}$$

then c *is strictly less than 1 and*

$$\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) \leq c^{n_1+n_2-1} \tag{12}$$

Theorem 4.5. *Assume a Full House Model in Definition 2.1 and additionally specify that* $Y_i \sim N(\mu_i, \sigma_i^2)$, *and suppose that the system inclusion mechanism has the property that* $1_i(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}) = 1$ *for* $1 = 1, 2$. *Let* $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ *where* $\mathbf{\Sigma}$ *is as defined in* (6) *and* (7). *Suppose that* $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\sigma}_1$ *and* $\hat{\sigma}_2$ *are, respectively, root sample-size consistent estimators of* μ_1, μ_2, σ_1 *and* σ_2. *Then*

$$\begin{aligned} & \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ & = O_P(n_1^{\frac{3}{2}} n_2 c^{n_1+n_2-3}) \end{aligned}$$

where c *is defined in* (11). *The main issue is the definition of* $w_{n,max}$. *This quantity is stated in complete generality as a function with arbitrary entries. However, its main use is when the entries take a specific form as stated in the proof of this theorem. The general version is not well-defined as* $w_{n,max}$ *is a function of the entries themselves. The specific version of* $w_{n,max}$ *with either population parameters or their estimators plugged-in is well-defined. More work is needed to determine what happens to* $w_{n,max}$ *as* $n \to \infty$ *and the corresponding* $\hat{w}_{n,max}$ *as* $n \to \infty$. *I do not expect the argument to be deal-breaking as the estimators exhibit nice convergence properties and, with population parameters defined within the interior of the parameter space, will be away from the boundary as* $n \to \infty$. *Also note that your Theorem conclusion is that the difference in probabilities is* $O_P(n_1^{\frac{3}{2}} n_2 w_{n_1+n_2-2,max}^{n_1+n_2-3})$ *whereas the end of the proof states that the difference goes to 0. These are two different things.*

Proof. Also, noticed that when $n \geq m$, $w_{n,max} \geq w_{m,max}$.

Let $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\sigma}_1$, and $\hat{\sigma}_2$ be consistent estimators of the population parameters μ_1, μ_2, σ_1, and σ_2, respectively, as defined in Section 4.2. By replacing the population parameters in $\mathbf{w}$ and $\mathbf{\Sigma}$ with these estimators, we obtain the sample-level analogs $\hat{\mathbf{w}}$ and $\hat{\mathbf{\Sigma}}$. Specifically, we define:

$$\hat{\mathbf{w}} = \left(\frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, \ldots, \frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, 0, \ldots, 0 \right) \in \mathbb{R}^{n_1+n_2-1},$$

where the first n_1 entries correspond to the standardized mean difference, and the remaining $n_2 - 1$ entries are zero, reflecting the structure introduced in Section 4.2.

The estimated covariance matrix $\hat{\mathbf{\Sigma}}$ has the block form:

$$\hat{\mathbf{\Sigma}} = \begin{bmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{bmatrix},$$

where each submatrix is obtained by replacing the population parameters in $\mathbf{\Sigma}$ with their respective sample estimates. These submatrices retain the same structural forms as defined in Section 4.2.

Based on the Equation (8), we can rewrite

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ &= n_2(\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma)) \end{aligned}$$

Consider the term $\Phi(\hat{\mathbf{w}}, \hat{\mathbf{\Sigma}}) - \Phi(\mathbf{w}, \mathbf{\Sigma})$. We perform a Taylor expansion of this expression with respect to the parameters μ_1, μ_2, σ_1, and σ_2.

$$\begin{aligned} \Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx &\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\ &+ \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2) \end{aligned}$$

Then for the first derivative term $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$:

$$\begin{aligned} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_1} \\ &= -\frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \end{aligned}$$

By applying Leibniz rule and without loss of generality, we compute the $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_1}$

$$\frac{\partial \Phi(\mathbf{w}; \mathbf{\Sigma})}{\partial w_1} = \frac{\partial}{\partial w_1} \int_{-\infty}^{w_1} \cdots \int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{w}}(z_1, \ldots, z_{n_1+n_2-1}) dz_{n_1+n_2-1} \ldots dz_1$$

$$
\begin{aligned}
&= \int_{-\infty}^{w_2} \cdots \int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(w_1, z_2, \ldots, z_{n_1+n_2-1}) dz_{n_1+n_2-1} \ldots dz_2 \\
&= \int_{-\infty}^{w_2} \cdots \int_{-\infty}^{w_{n_1+n_2-1}} f_{W_1}(w_1) \cdot f_{\tilde{\mathbf{W}}}(\tilde{w}) dz_{n_1+n_2-1} \ldots dz_2 \\
&= \phi(w_1)\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})
\end{aligned}
$$

where $\phi(w_1)$ is the probability density function of univariate normal distribution with $\mu = 0$ and $\sigma^2 = \frac{\sigma_2^2}{\sigma_1^2} + 1$, $\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$ is the conditional cumulative distribution function of the remaining components given $W_1 = w_1$. By applying Schur complement [Zhang, 2006],

$$
\tilde{\mathbf{w}} = (w_2, \ldots, w_{n_1+n_2-1}) - (\frac{\sigma_2^2}{\sigma_1^2}, \ldots, \frac{\sigma_2^2}{\sigma_1^2}, \frac{\sigma_2}{\sigma_1}, \ldots, \frac{\sigma_2}{\sigma_1}) * (\frac{\sigma_2^2}{\sigma_1^2} + 1)^{-1} * (w_1)
$$

$\tilde{\mathbf{w}}$ is a vector with the first $n_1 - 1$ elements is $\frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\mu_2 - \mu_1}{\sigma_1} \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$ and the last $n_2 - 1$ elements are $-\frac{(\mu_2 - \mu_1)\sigma_2}{\sigma_1^2 + \sigma_2^2}$.

$$
\tilde{\mathbf{\Sigma}} = \mathbf{\Sigma}_{-1,-1} - \mathbf{\Sigma}_{-1,1} \mathbf{\Sigma}_{1,1}^{-1} \mathbf{\Sigma}_{1,-1}
$$

where $\mathbf{\Sigma}_{-1,-1}$ denotes the submatrix of $\mathbf{\Sigma}$ obtained by removing the first row and the first column, $\mathbf{\Sigma}_{-1,1}$ denotes the column vector obtained by removing the first element from the first column of $\mathbf{\Sigma}$, $\mathbf{\Sigma}_{1,-1}$ denotes the row vector obtained by removing the first element from the first row of $\mathbf{\Sigma}$, and $\mathbf{\Sigma}_{1,1}$ denotes the top-left scalar element of $\mathbf{\Sigma}$.

Let denote the form of the matrix $\tilde{\mathbf{\Sigma}}$:

$$
\tilde{\mathbf{\Sigma}} = \begin{bmatrix} \tilde{\Sigma}_{11} & \tilde{\Sigma}_{12} \\ \tilde{\Sigma}_{21} & \tilde{\Sigma}_{22} \end{bmatrix}
$$

$$
\tilde{\Sigma}_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}_{(n_1-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2} + 1} \begin{bmatrix} \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4} \end{bmatrix}_{(n_1-1)\times(n_1-1)}
$$

$$
= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_1-1)\times(n_1-1)}
$$

$$\tilde{\Sigma}_{12}^T = \tilde{\Sigma}_{21} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

$$= \begin{bmatrix} 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \dots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \dots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Therefore, $\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$ is finite and by applying Lemma 4.4, we get

$$\begin{aligned} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1} &= -\frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \\ &= -\frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) \\ &\leq O_P(n_1 c^{n_1+n_2-1}) \end{aligned}$$

By applying the similar method used for $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$, we have

$$\begin{aligned}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2} &= \sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}\cdot\frac{\partial w_i}{\partial\mu_2}\\ &= \frac{1}{\sigma_1}\sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}\\ &= \frac{n_1}{\sigma_1}\cdot\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})\\ &= OP(n_1 c^{n_1+n_2-1})\end{aligned}$$

Notice that $\Phi(\mathbf{w},\Sigma)$ has two parameters $\mathbf{w}$, and Σ involving σ_1, then using the Equation (137) in Petersen et al. [2008] and the similar method used for $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, we have

$$\begin{aligned}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1} &= \sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}\cdot\frac{\partial w_i}{\partial\sigma_1}+\mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})\\ &= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})+\mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})\end{aligned}$$

Before computing $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}$, we define the $\phi(\mathbf{x})$ be the probability density function of $\Phi(\mathbf{w},\Sigma)$, which is

$$\phi(\mathbf{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma)}}\exp\left[-\frac{1}{2}(\mathbf{x})^T\Sigma^{-1}(\mathbf{x})\right]$$

then

$$\log(\phi(\mathbf{x})) = -\frac{1}{2}(\log(\det(2\pi\Sigma))+\mathbf{x}^t\Sigma^{-1}\mathbf{x})$$

then using the Equation (57) and (61) in Petersen et al. [2008], we have

$$\begin{aligned}\frac{\partial\log(\phi(\mathbf{x}))}{\partial\Sigma} &= -\frac{1}{2}(\frac{\partial\log(\det(2\pi\Sigma))}{\partial\Sigma}+\frac{\partial\mathbf{x}^t\Sigma^{-1}\mathbf{x}}{\partial\Sigma})\\ &= -\frac{1}{2}(\Sigma^{-1}-\Sigma^{-1}\mathbf{x}\mathbf{x}^t\Sigma^{-1})\end{aligned}$$

Then for the term $\mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})$ is:

$$\begin{aligned}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma} &= \frac{\partial\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})d\mathbf{x}}{\partial\Sigma}\\ &= \int_{-\infty}^{\mathbf{w}}\frac{\partial\phi(\mathbf{x})}{\partial\Sigma}d\mathbf{x}\\ &= \int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\frac{\partial\log(\phi(\mathbf{x}))}{\partial\Sigma}d\mathbf{x}\end{aligned}$$

$$= -\frac{1}{2}\Sigma^{-1}\Phi(\mathbf{w},\Sigma) + \frac{1}{2}\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x})\cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}$$
$$= \frac{1}{2}\Sigma^{-1}\left(\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x})\cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w},\Sigma)\right)$$

Then

$$\begin{aligned}\operatorname{tr}\left(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1}\right) &= \operatorname{tr}\left(\frac{1}{2}\Sigma^{-1}\left(\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x})\cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w},\Sigma)\right)\frac{\partial\Sigma}{\partial\sigma_1}\right)\\ &= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x})\cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)\end{aligned}$$

To compute the $\operatorname{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})$, we firstly compute the $\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)$. Let define $\frac{\partial\Sigma}{\partial\sigma_1}$ be:

$$\frac{\partial\Sigma}{\partial\sigma_1} = \begin{bmatrix}\frac{\partial\Sigma_{11}}{\partial\sigma_1} & \frac{\partial\Sigma_{12}}{\partial\sigma_1}\\ \frac{\partial\Sigma_{21}}{\partial\sigma_1} & \frac{\partial\Sigma_{22}}{\partial\sigma_1}\end{bmatrix} = \begin{bmatrix}A_{11} & A_{12}\\ A_{21} & A_{22}\end{bmatrix}$$

$$A_{11} = \left[-\frac{2\sigma_2^2}{\sigma_1^3}\right]_{n_1\times n_1}, A_{12}^T = A_{21} = \left[-\frac{\sigma_2}{\sigma_1^2}\right]_{(n_2-1)\times n_1}, A_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

and by applying Schur complement [Zhang, 2006], we have

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{bmatrix}^{-1} = \begin{bmatrix}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & -(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}\\ -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & \Sigma_{22}^{-1}+\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}\end{bmatrix}$$

where Σ_{11}, Σ_{12}, Σ_{21}, and Σ_{22} are defined in Equation (6).
By applying Equation (10), we have

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix}B_{11} & B_{12}\\ B_{21} & B_{22}\end{bmatrix}$$

$$B_{11} = (\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$$
$$= \left(\begin{bmatrix}\frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1\end{bmatrix} - \begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}\begin{bmatrix}2 & 1 & \dots & 1\\ 1 & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ 1 & \dots & 1 & 2\end{bmatrix}^{-1}\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}\right)^{-1}$$

$$
\begin{aligned}
&= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1} \\
&= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - (1-\frac{1}{n_2}) \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix} \right)^{-1} \\
&= \begin{bmatrix} \frac{\sigma_2^2}{n_2\sigma_1^2}+1 & \frac{\sigma_2^2}{n_2\sigma_1^2} & \dots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \dots & \frac{\sigma_2^2}{n_2\sigma_1^2} & \frac{\sigma_2^2}{n_2\sigma_1^2}+1 \end{bmatrix}^{-1} = \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}_{n_1\times n_1}
\end{aligned}
$$

$$
\begin{aligned}
B_{12} &= -(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}
\end{aligned}
$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{n_1 \times (n_2-1)}$$

$B_{21} = -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$

$$= -\begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}$$

$$= -\begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{(n_2-1)\times n_1}$$

$B_{22} = \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$

$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$
= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}
$$

$$
= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}
$$

$$
= \begin{bmatrix} 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}
$$

Then by applying Lemma 4.4, we have

$$\Phi(\mathbf{w}, \Sigma) \leq c^{n_1+n_2-1}$$

Therefore,

$$\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_1} \leq \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}c^{n_1+n_2-1}$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_1}$, we have

$$
\begin{aligned}
&\operatorname{tr}(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}A_{11} + B_{12}A_{21}) + tr(B_{21}A_{12} + B_{22}A_{22}) \\
&= n_1\left(\frac{-\frac{2\sigma_2^2}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2}) + \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{\sigma_2}{\sigma_1^2}(n_2-1)\right)
\end{aligned}
$$

$$+ (n_2 - 1) \left(n_1 \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} \frac{\sigma_2}{\sigma_1^2} \right)$$
$$= -2 \frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}$$

Now we have

$$\mathrm{tr}(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_1}) \leq 2 \frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} c^{n_1+n_2-1}$$

Now we compute the term $\mathrm{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1}\right)$. Let rewrite the $\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}$ into matrix form:

$$\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix}$$

Define $\mathbf{w}_{-i}$ as the vector $\mathbf{w}$ with the ith component removed, and $\mathbf{w}_{-(i,j)}$ as the vector $\mathbf{w}$ with the ith and j th components removed. Similarly, $\mathbf{x}_{-i}$ and $\mathbf{x}_{-(i,j)}$ are defined in the same manner.

Then

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i^2 d\mathbf{x} &= \int_{-\infty}^{w_i} \left(\int_{-\infty}^{\mathbf{w}_{-i}} \phi(\mathbf{x}_{-i}|x_i) d\mathbf{x}_{-i} \right) \; x_i^2 \; \phi(x_i) dx_i \\ &\leq c^{n_1+n_2-3} \cdot Var(X_i) \end{aligned}$$

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i x_j d\mathbf{x} &= \int_{-\infty}^{w_i} \int_{-\infty}^{w_j} \left(\int_{-\infty}^{\mathbf{w}_{-(i,j)}} \phi(\mathbf{x}_{-(i,j)}|x_i, x_j) d\mathbf{x}_{-(i,j)} \right) \; x_i x_j \; \phi(x_i, x_j) dx_i dx_j \\ &\leq c^{n_1+n_2-3} \cdot Cov(X_i, X_j) \end{aligned}$$

Replace $Var(X_i)$ and $Cov(X_i, X_j)$ using the elements of Σ, we have

$$\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix}$$

$$\leq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

where

$$S_{11} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) \end{bmatrix}_{n_1 \times n_1}$$

$$S_{12}^T = S_{21} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \end{bmatrix}_{(n_2-1)\times n_1}$$

$$S_{22} = \begin{bmatrix} 2c^{n_1+n_2-3} & c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} & 2c^{n_1+n_2-3} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Therefore,

$$\begin{aligned}
&\operatorname{tr}\left(\Sigma^{-1}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11}S_{11}+B_{12}S_{21} & B_{11}S_{12}+B_{12}S_{22} \\ B_{21}S_{11}+B_{22}S_{21} & B_{21}S_{12}+B_{22}S_{22} \end{bmatrix}\begin{bmatrix} B_{11}A_{11}+B_{12}A_{21} & B_{11}A_{12}+B_{12}A_{22} \\ B_{21}A_{11}+B_{22}A_{21} & B_{21}A_{12}+B_{22}A_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}((B_{11}S_{11}+B_{12}S_{21})(B_{11}A_{11}+B_{12}A_{21})+(B_{11}S_{12}+B_{12}S_{22})(B_{21}A_{11}+B_{22}A_{21})) \\
&+ \operatorname{tr}((B_{21}S_{11}+B_{22}S_{21})(B_{11}A_{12}+B_{12}A_{22})+(B_{21}S_{12}+B_{22}S_{22})(B_{21}A_{12}+B_{22}A_{22})) \\
&= -\frac{\frac{n_1\sigma_2^2}{\sigma_2^3}(\frac{2n_1\sigma_2^2}{\sigma_2^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{n_1\sigma_2^2}{\sigma_2^3}\frac{n_2-1}{n_2}(\frac{n_1\sigma_2^2}{\sigma_2^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{2n_1^2\sigma_2^4}{\sigma_2^5}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3} \\
&- \frac{\frac{n_1^2\sigma_2^4}{\sigma_2^5}(\frac{n_2-1}{n_2})^2}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{2n_1^2\sigma_2^4}{\sigma_2^5}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}-1)c^{n_1+n_2-3} \\
&- \frac{\frac{2n_1^2\sigma_2^4}{\sigma_2^5}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(1-\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)})c^{n_1+n_2-3}
\end{aligned}$$

$$+\frac{\frac{n_1\sigma_2^2}{\sigma_2^3}\frac{n_2-1}{n_2}(n_1\frac{\sigma_2^2}{\sigma_1^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}-1)c^{n_1+n_2-3}$$
$$-\frac{\frac{n_1^2\sigma_2^4}{\sigma_2^5}(\frac{n_2-1}{n_2})^2}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}+\frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(1-\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)})c^{n_1+n_2-3}$$
$$=O_P(n_1^2c^{n_1+n_2-3})$$

Then

$$\begin{aligned}\mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})&=\frac{1}{2}\,\mathrm{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right)-\frac{1}{2}\,\mathrm{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)\\&=O_p(n_1^2c^{n_1+n_2-3})\end{aligned}$$

By applying the similar method used for $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, we have

$$\begin{aligned}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1}&=-\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})+\mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})\\&=-\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})+O_p(n_1^2c^{n_1+n_2-3})\\&=O_P(n_1^2c^{n_1+n_2-3})\end{aligned}$$

Similarly, we compute $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}$,

$$\begin{aligned}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}&=\mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_2})\\&=\frac{1}{2}\,\mathrm{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}\right)-\frac{1}{2}\,\mathrm{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_2}\right)\end{aligned}$$

To compute $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}$, firstly we have

$$\begin{aligned}\frac{\partial\Sigma}{\partial\sigma_2}&=\begin{bmatrix}\frac{\partial\Sigma_{11}}{\partial\sigma_1}&\frac{\partial\Sigma_{12}}{\partial\sigma_1}\\\frac{\partial\Sigma_{21}}{\partial\sigma_1}&\frac{\partial\Sigma_{22}}{\partial\sigma_1}\end{bmatrix}\\&=\begin{bmatrix}C_{11}&C_{12}\\C_{21}&C_{22}\end{bmatrix}\end{aligned}$$

$$C_{11}=\left[\frac{2\sigma_2}{\sigma_1^2}\right]_{n_1\times n_1},C_{12}^T=C_{21}=\left[\frac{1}{\sigma_1}\right]_{(n_2-1)\times n_1},C_{22}=\left[0\right]_{(n_2-1)\times(n_2-1)}$$

Then by applying Lemma 4.4, we have

$$\Phi(\mathbf{w}, \Sigma) \leq c^{n_1+n_2-1}$$

Then

$$\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_2} \leq \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}c^{n_1+n_2-1}$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_2}$, we have

$$\begin{aligned}
&\operatorname{tr}(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}C_{11} + B_{12}C_{21}) + tr(B_{21}C_{12} + B_{22}C_{22}) \\
&= n_1\left(\frac{\frac{2\sigma_2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2})\right) - \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{1}{\sigma_1}n_1(n_2-1) \\
&- (n_2-1)\left(n_1\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{1}{\sigma_1}\right) \\
&= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}
\end{aligned}$$

Now we have

$$\operatorname{tr}(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_2}) \leq \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-1}$$

$$\begin{aligned}
&\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) \leq \operatorname{tr}\left(\Sigma^{-1}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}((B_{11}S_{11} + B_{12}S_{21})(B_{11}C_{11} + B_{12}C_{21}) + (B_{11}S_{12} + B_{12}S_{22})(B_{21}C_{11} + B_{22}C_{21})) \\
&+ \operatorname{tr}((B_{21}S_{11} + B_{22}S_{21})(B_{11}C_{12} + B_{12}C_{22}) + (B_{21}S_{12} + B_{22}S_{22})(B_{21}C_{12} + B_{22}C_{22})) \\
&= \frac{\frac{n_1\sigma_2}{\sigma_2^2}(\frac{2n_1\sigma_2^2}{\sigma_2^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3} - \frac{\frac{n_1\sigma_2}{\sigma_2^2}\frac{n_2-1}{n_2}(\frac{n_1\sigma_2^2}{\sigma_2^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3} - \frac{\frac{2n_1^2\sigma_2^3}{\sigma_2^4}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}
\end{aligned}$$

$$+\frac{\frac{n_1^2\sigma_2^3}{\sigma_2^4}(\frac{n_2-1}{n_2})^2}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}-\frac{\frac{2n_1^2\sigma_2^3}{\sigma_2^4}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}-\frac{\frac{n_1(n_2-1)\sigma_2}{\sigma_2^2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}-1)c^{n_1+n_2-3}$$
$$+\frac{\frac{2n_1^2\sigma_2^3}{\sigma_2^4}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}-\frac{\frac{n_1(n_2-1)\sigma_2}{\sigma_2^2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(1-\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)})c^{n_1+n_2-3}-\frac{\frac{n_1\sigma_2}{\sigma_2^2}\frac{n_2-1}{n_2}(n_1\frac{\sigma_2^2}{\sigma_1^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}$$
$$-\frac{\frac{n_1(n_2-1)\sigma_2}{\sigma_2^2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}-1)c^{n_1+n_2-3}+\frac{\frac{n_1^2\sigma_2^3}{\sigma_2^4}(\frac{n_2-1}{n_2})^2}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}c^{n_1+n_2-3}$$
$$-\frac{\frac{n_1(n_2-1)\sigma_2}{\sigma_2^2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(1-\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)})c^{n_1+n_2-3}$$
$$=O_P(n_1^2c^{n_1+n_2-3})$$

Then we put $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2}$, $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1}$, and $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}$ together and based on the definition of root sample-size consistent estimator,

$$|\hat{\mu}_1-\mu_1|=O_p(n_1^{-1/2}),$$

$$|\hat{\mu}_2-\mu_2|=O_p(n_2^{-1/2}),$$

$$|\hat{\sigma}_1-\sigma_1|=O_p(n_1^{-1/2}),$$

$$|\hat{\sigma}_2-\sigma_2|=O_p(n_2^{-1/2}),$$

we have

$$\begin{aligned}\Phi(\hat{\mathbf{w}},\hat{\Sigma})-\Phi(\mathbf{w},\Sigma)\approx&\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}(\hat{\mu_1}-\mu_1)+\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2}(\hat{\mu_2}-\mu_2)+\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1}(\hat{\sigma_1}-\sigma_1)\\&+\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}(\hat{\sigma_2}-\sigma_2)\\c=&O_P(n_1^{\frac{3}{2}}c^{n_1+n_2-3})\end{aligned}$$

Finally,

$$\begin{aligned}&\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)},Y_{1,(n_1)}<Y_{2,(n_2)}|\hat{\mu_1}.\hat{\mu_2},\hat{\sigma_1},\hat{\sigma_2}\right)-\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)},Y_{1,(n_1)}<Y_{2,(n_2)}|\mu_1.\mu_2,\sigma_1,\sigma_2\right)\\&=O_P(n_1^{\frac{3}{2}}n_2c^{n_1+n_2-3})\end{aligned}$$

□

5 Examples

5.1 Sensitivity Analysis for Theorem 3.4

In this section, we conduct a sensitivity analysis to evaluate the influence of the parameters N_1, N_2, n_1, and n_2 in Theorem 3.4.

Under the setting of Theorem 3.4, we assume $X_1 = X_2 = X$, $Y_1 = Y_2 = Y$, where $X \sim \text{Pareto}(\alpha)$ and $Y \sim \mathcal{N}(0, 1)$. Without loss of generality, we set $j = n_1 - 1$ and $k = n_2 - 1$, corresponding to the selection of the second-best performer in the paradox probability, as defined in Definition 3.1.

In each simulation, we randomly sample n_1 and n_2 observations from the distribution of Y, and compute their latent aptitudes using Equation (1). Based on these values, we evaluate the paradox probability according to Definition 3.1. We repeat this simulation 10000 times and report the average paradox probability.

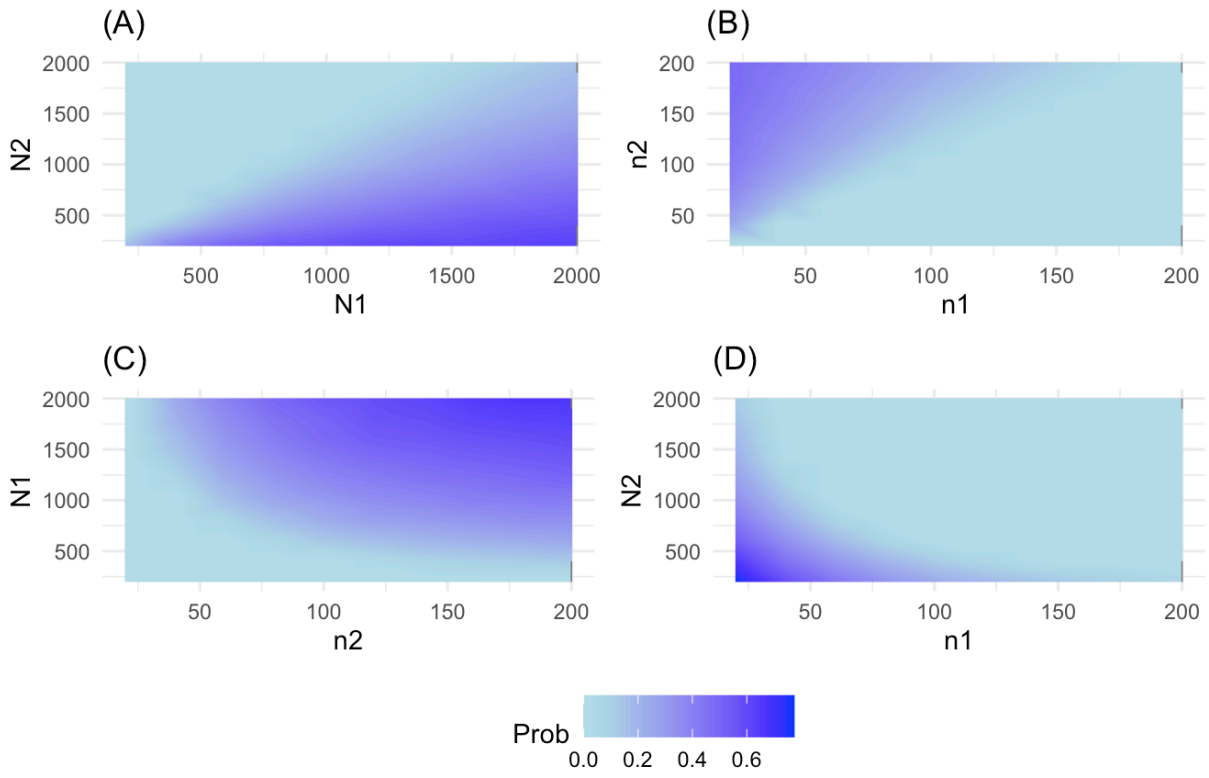


Figure 1: Sensitivity Analysis for Paradox Probability

Figure 1 presents the results of the sensitivity analysis for the paradox probability described in Theorem 3.4. In Figure 1 (A), we fix $n_1 = 40$ and $n_2 = 60$, and vary N_1 and N_2 from 200 to 2000. In Figure 1 (B), we fix $N_1 = 10^3$ and $N_2 = 10^3$, and vary n_1 and n_2 from 20 to 200. In Figure 1 (C), we fix $n_1 = 40$ and $N_2 = 10^3$, and vary n_1 from 20 to 200 and N_2 from 200 to 2000. In Figure 1 (D), we fix $N_1 = 10^3$ and $n_2 = 60$, and vary N_1 from 200 to 2000 and n_2 from 20 to 200.

As shown in Figure 1 (A), a small N_2 and large N_1 increase the paradox probability by making $\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}\right)$ approach 1. Similarly, in Figure 1 (B), a small n_1 combined with a large n_2 leads to a higher paradox probability, since it drives $\mathrm{P}\left(Y_{1,(j)} < Y_{2,(k)}\right)$ towards 1.

We also investigate the effect of the ratio between N_2 and N_1 on the paradox probability. Let $c = \frac{N_2}{N_1}$, and vary c from 0.1 to 1 in increments of 0.1. Without loss of generality, we set $j = n_1 - 1$ and $k = n_2 - 1$, corresponding to the selection of the second-best performer in the paradox probability, as defined in Definition 3.1.

In each simulation, we fix $n_1 = 60$ and $n_2 = 40$, and vary N_1 from 100 to 2000. We repeat this simulation 10000 times and report the average paradox probability.

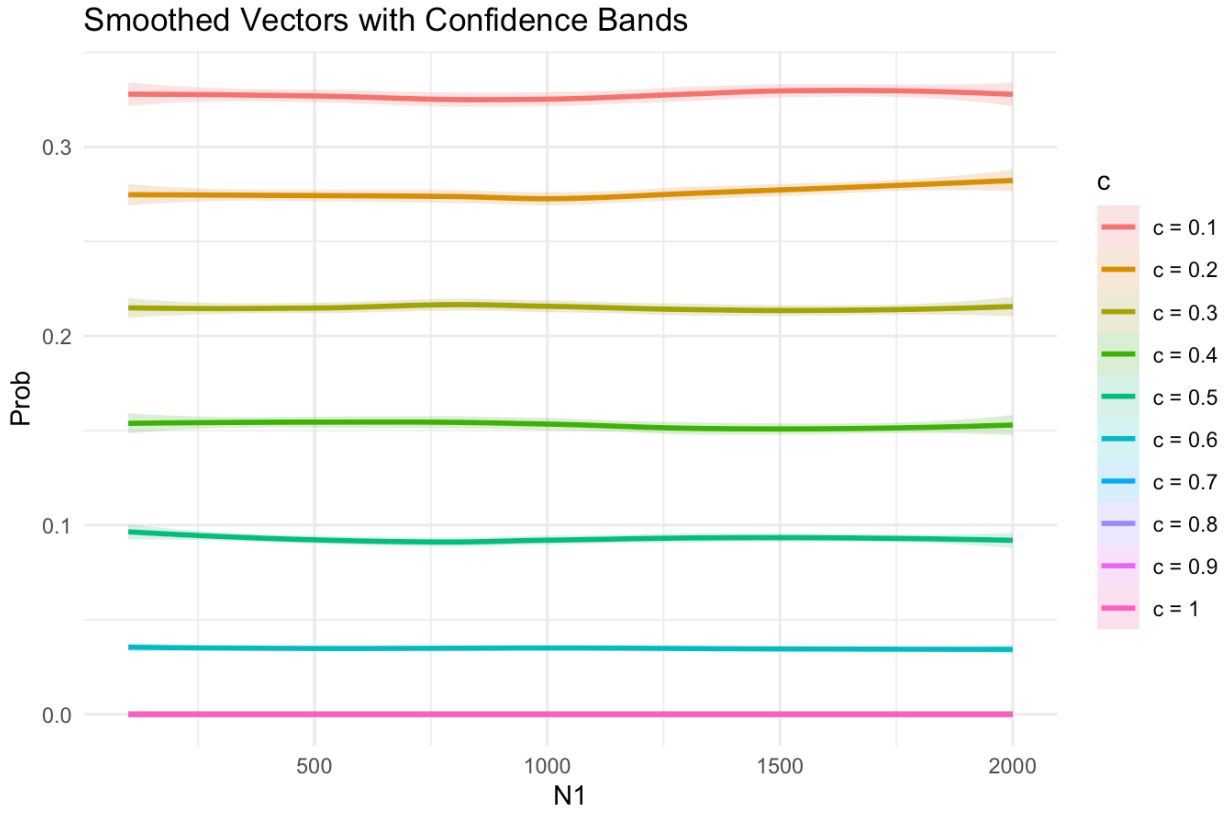


Figure 2: Effect of the ratio between N_2 and N_1 on the paradox probability

Figure 2 presents the results illustrating the effect of the ratio between N_2 and N_1 on the paradox probability under different settings. As shown in the figure, the paradox probability decreases as c increases, which is reasonable since $\mathbb{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}\right)$ decreases as N_2 increases.

Figure 2 also demonstrates that the paradox probability remains constant when the ratio between N_2 and N_1 is fixed. This implies that it is unnecessary to focus on the exact values of N_1 and N_2 when computing the paradox probability; rather, the ratio N_2/N_1 alone determines its value.

5.2 Recursive Full House Models

In this section we will investigate a simple recursive Full House Models in which Full House Models are repeatedly applied to individuals as they progress through iterations of some generic process. The general setup is that, in the first round, latent aptitude $X_j \overset{iid}{\sim} F_X$, $j = 1, \ldots, N$ will be generated and then a label will be randomly assigned to each individual. There will be m labels and each label will designate the outcome distribution. $F_{Y_{i,j}}$, $i = 1, \ldots, n$, $j = 1, \ldots, n_i$. Then outcomes will be realized as

$$Y_{i,(r_{i,(j)})} = F^{-1}_{Y_{i,j}} \circ I^{-1}_{r_{i,(j)}, n_i - r_{i,(j)}+1} \circ I_{j,N-j+1} \circ F_X(X_{(j)}), \tag{13}$$

where $r_{i,(j)}$ is the rank of $X_{(j)}$ among the n_i latent aptitudes also assigned label i.

In the next iteration, all observations will be combined and then sorted in descending order by $Y_{i,(j)}$. The labels will be assigned in descending order, the largest n_m values of $Y_{i,(j)}$ are assigned label m, the next largest n_{m-1} values of $Y_{i,(j)}$ are assigned label $m-1$, and so on. We will setup these simulation so that Y_i stochastically dominates Y_j whenever $i > j$ so that higher outcomes are associated with higher labels. Then new Y values are generated according to (13). The process will repeat until a pre-assigned stopping rule is satisfied. Note that this simulation setup assumes an immutable latent aptitude.

5.2.1 Stop when highest latent aptitude is included in the highest-performing group

Here we consider a recursive Full House Modeling setup described in Section 5.2 in which the recursion stops at the first iteration when $X_{(N)}$ is assigned label m. Here we specify that $N = 100,000$, $m = 200$, $n_i = 500$, $i = 1, \ldots, m$. We will assume that $X_j \overset{iid}{\sim}$ Pareto(1.16), $j = 1, \ldots, N$. And we will assume that $Y_{i,j} \overset{iid}{\sim} N(\mu_i, \sigma_i)$, $j = 1, \ldots, n_i$ and $i = 1, \ldots, m$. We will specify that the μ_is as a set of 200 equally spaced values between 60 and 80 with μ_i increasing in i, ie $\mu_i = 60 - 20/199 + 20i/199$, $i = 1, \ldots, m$. We will fix $\sigma_i = \sigma$ and consider $\sigma \in \{0.5, 1, 2\}$. These simulation will be repeated 500 times.

To illustrate the procedure, consider a simplified scenario in which all $X_{(j)}$ values are realized at their median value. With this simplification, $Y_{i,(r_{i,(j)})}$ in (13) is approximately $F^{-1}_{Y_{i,j}}((r_{i,(j)} - 1/3)/(n_i + 1/3))$ [Kerman, 2011]. And we have that $Y_{i,(n_i)} - Y_{i+1,(n_{i+1}-1)} > \mu_{i+1} - \mu_i$ for all considered values of σ. The individual with latent ability $X_{(N)}$ will consistently attain the value $Y_{i,(n_i)}$, regardless their assigned group. Consequently, this individual's position among the m groups will improve as the simulation advances.

Results can be seen in Figure 3. These Monte Carlo simulations reveal that it typically takes several rounds before the individual with the top aptitude belongs to the group that allows for the highest performances. On average, with σ set to 0.5, 1, and 2, it takes 2.61, 4.19, and 7.46 rounds, respectively, for the individual with the top aptitude to join the highest-performing group (respective standard errors: 0.05, 0.09, 0.18).

These findings resonate with several empirical studies examining bias and distortion in real-world evaluative systems. The GRADE algorithm developed by Waters and Miikkulainen [2014] demonstrates how admission systems can bypass nuanced information (e.g., personal statements) in favor of machine-learned proxies, potentially reinforcing existing filters. Similarly, Alvero et al. [2020] show that seemingly innocuous features (such as the

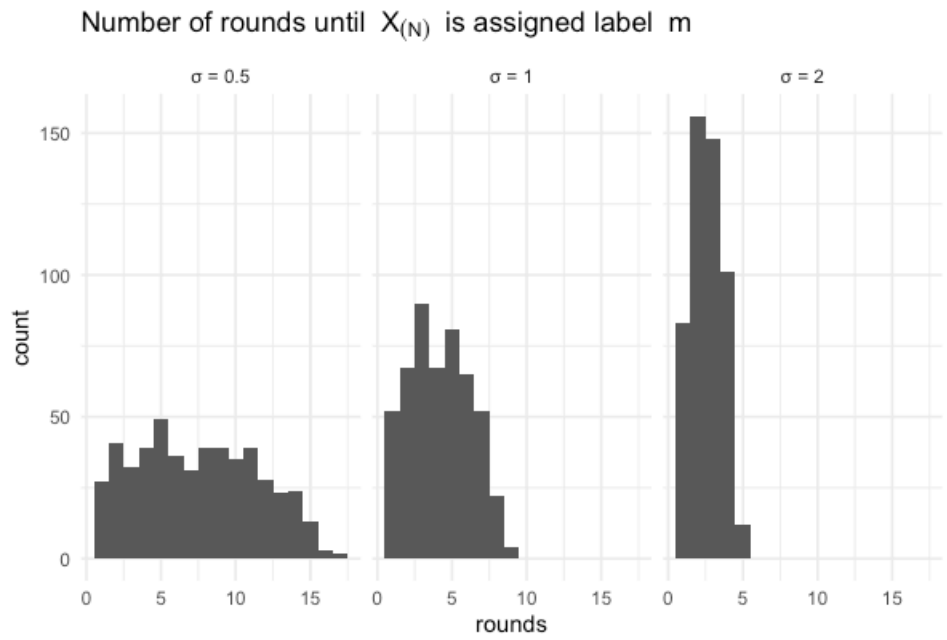


Figure 3: The number of rounds that it takes for $X_{(N)}$ to be assigned label m in a recursive Full House Modeling framework in Section 5.2 with specifications in Section 5.2.1.

word baseball) can serve as strong demographic signals. This raises concerns about indirect inference. A broader review by Baker and Hawn [2022] warns that algorithmic systems may harbor unknown unknowns, especially in global or under-examined subpopulations. Finally, the meta-analyses in Walton and Spencer [2009] provide direct evidence that outcome-based measures (e.g., grades, test scores) can systematically understate the aptitude of marginalized groups due to identity-based threat.

These phenomena of latent talent being misclassified, sorted, and repeatedly filtered through context-specific outcome layers is precisely what this presentation of recursive Full House Modeling aims to formalize and study. At any stage in the selection process, you cannot be sure that the highest marks correspond to individuals with the highest ability. While the setting we consider is stylized and assumes fixed latent aptitude, it serves to illustrate how structural features of evaluative systems, not just randomness, can delay or obscure the recognition of true potential.

5.3 Batting Averages in Major League Baseball

We now apply Theorem 4.3 and the reliability guarantees in Theorem 4.5 to evaluate the contextual extremeness of observed outcomes in a real-world system where the underlying distribution of aptitude is not directly observable and evolves over time. Major League Baseball (MLB) offers a canonical example of such a system. Here, player skill (latent batting aptitude) is inferred only indirectly through season-level observed outcomes (batting averages), which are shaped by changing rules, population structures, and environmental variance.

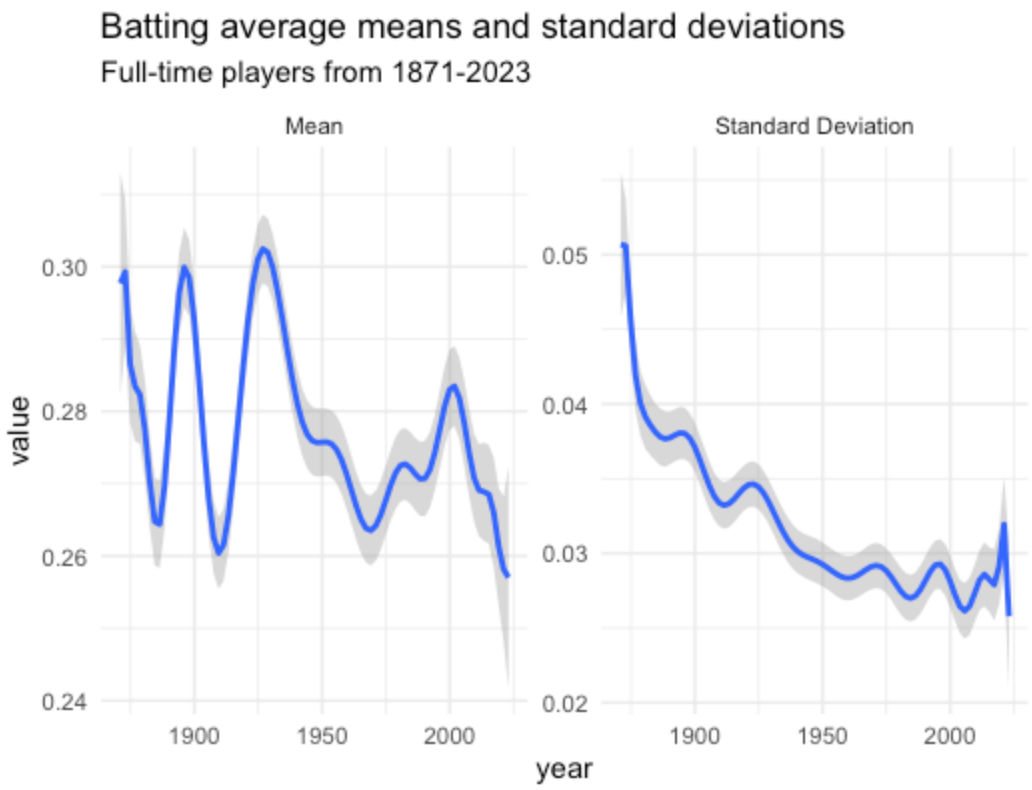


Figure 4: Seasonal means and standard deviations of batting averages for full-time players from 1871-2023. Smoothing is applied to both trends.

In particular, MLB introduced a series of rule changes in 2023 intended to increase in-game action, including a restriction on defensive positioning designed to raise batting averages on balls in play (BABIP) [Castrovince, 2023].

First, the 2023 mean batting average for full-time players increased by 0.0057 (corresponding to an additional hit about every four baseball games) with a reduction in variability. A two-sample tests for differences in batting averages among full-time players from the 2023 and 2022 seasons has a p-value of 0.039. Thus, the rule changes seemed to be slightly effective in yielding increased batting averages[1] Moreover, the 2023 MLB batting leader Luis Arraez had a 0.354 batting average which was the first batting average to exceed .350 since Josh Hamilton in 2010.

Now we turn to applying the results of Theorem 4.3 in this example to contextualize the effect of these rule changes. Here we assume that seasonal batting averages of full-time players are normally distributed. The true means and standard deviations of these distributions are unknown. Thus, we estimate them via plug-in sample mean and sample standard-deviation and invoke the reliability guarantees from Theorem 4.5. Figure 4 presents smoothed estimates of the mean and standard deviation of batting averages for full-time players from 1871 to 2023. Notably, both the average and variability of batting averages

[1]Technically, Castrovince [2023] referred to MLB's desire to increased batting averages on balls in play (BABIP) while this analysis focused on batting averages. BABIP increased by 0.0061 from 2022 to 2023.

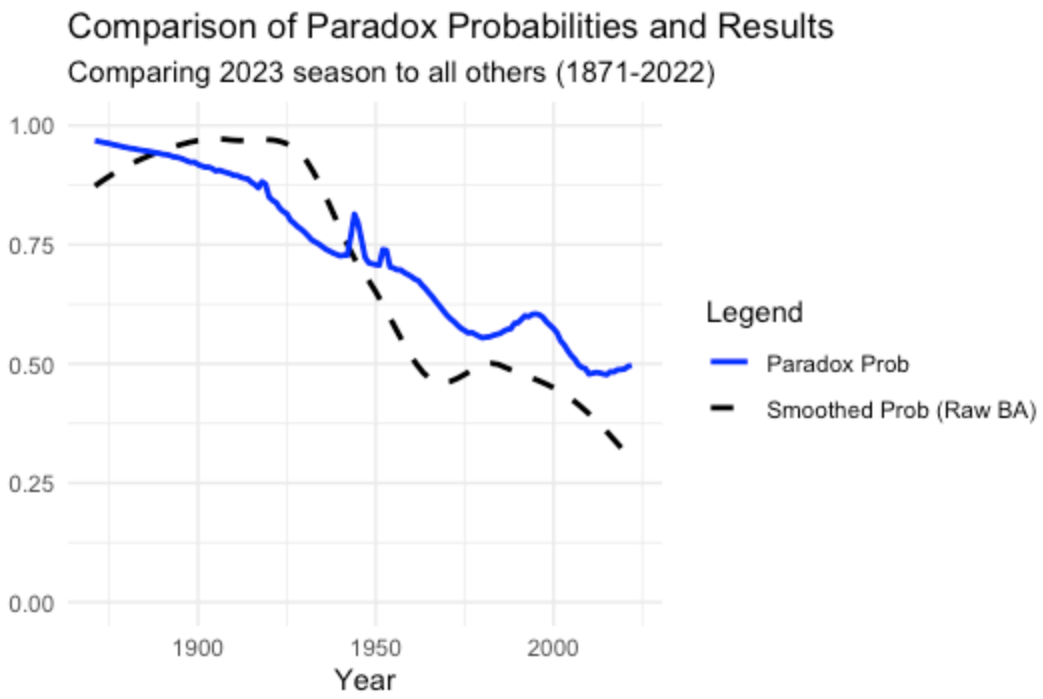


Figure 5: Comparison of paradox probabilities and empirical exceedance rates relative to the 2023 season, based on data from 1871-2022. The blue line shows estimated paradox probabilities via plug-in using Theorem 4.3. The black dashed line displays a smoothed empirical exceedance rate for Arraez's 2023 batting average.

in 2023 are among the lowest observed in the dataset, indicating a historically suppressed environment for batting averages of full-time batters.

Figure 5 displays paradox probabilities (8) for each season from 1871 to 2022, comparing them to the 2023 season in which Luis Arraez led MLB in batting average (treating 2023 as system 1 in the notation of Theorem 4.3). These probabilities are estimated using plug-in, sample means and standard deviations are substituted for population parameters. These estimators are $\sqrt{n}$-consistent so that the reliability results of Theorem 4.5 apply. For comparison, the figure also shows a smoothed empirical curve representing the proportion of seasons in which the maximum batting average exceeded Arraez's 2023 mark. We observe that the estimated paradox probabilities track closely with the empirical exceedance rate. Both quantities increase as we move backward in time, reinforcing the idea that earlier eras, with higher means, larger variances, and smaller populations, were more conducive to producing extreme individual batting averages, even when overall talent levels may have been lower.

Yan et al. [2025a] studied historical batting averages and era-adjusted batting averages through Full House Modeling for Luis Arraez's 2023 seasons and Jeff McNeil's MLB leading 2022 season in which he hit 0.326. They found that when one accounts for park and era effects then Arraez and McNeil had very similar performances, despite Arraez's large lead in raw batting average. These authors also found that assuming normality for batting averages

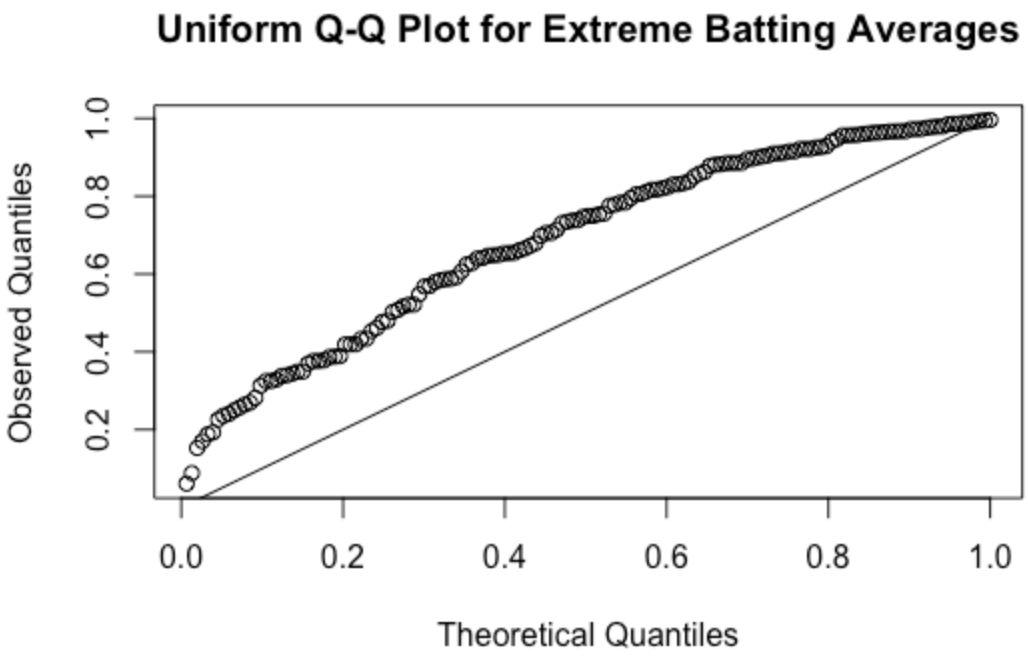


Figure 6: Uniform Q-Q plot for extreme batting averages defined as $\widehat{F}(Y_{(n_i),i})^{n_i}$ where i denotes the season, n_i denotes the number of full-time players in season i, $Y_{(n_i),i}$ denotes the highest batting average for full-time players in season i, and $\widehat{F}$ is the distribution function of the normal distribution with sample mean and standard deviation estimators substituted for the population parameters.

among full-time players might not be appropriate [Yan et al., 2025b]. Even so, we evaluate $\widehat{F}(Y_{(n_i),i})^{n_i}$, the observed percentile of the distribution for the maximum batting average $Y_{(n_i),i}$ in season i. These percentiles are not uniformly distributed as seen in Figure 6.

Interestingly, the percentile corresponding to Luis Arraez's 2023 MLB leading batting average is largely without recent precedent. See Table 1 for specifics. This table contains information from all seasons in which an MLB leading batter's extreme batting average percentile exceeds that of Luis Arraez in 2023. Notice that there is a significant drop off in the frequency of such seasons after Rogers Hornsby's modern day (post-1900) record 0.424 season from 1924. It is also interesting to note that the four most recent seasons in this table are from international players or players that would not have been allowed to play baseball had segregation continued (same with Rod Carew in 1977). Again, extreme performances were more common when the talent pool was more restricted and batting strategies and training methods were more variable, not when MLB expanded the number of teams [Gould, 1996, Eck, 2020a,b, Yan et al., 2025b]. This table also includes a column for p-values from a Shapiro-Wilk test [Shapiro and Wilk, 1965] on the normality of batting average for full-time players. These results support the findings in Yan et al. [2025b]. Moreover, the values $\widehat{F}(Y_{(n_i),i})^{n_i}$ do not follow a uniform distribution as indicated in Figure 6.

Despite the simplifying assumption of normality, the paradox probability analysis and associated diagnostics (Figures 5 and 6, Table 1) illustrate a key insight that under the empirical structure of 2023, which is characterized by low variance and high competition levels, Luis Arraez's 0.354 batting average represents an extremely rare achievement. Yet paradoxically, it appears modest when compared to higher averages in historically weaker environments.

This demonstrates a central tenet of Full House Modeling: observed records are shaped as much by the structure of the evaluation system as by individual aptitude. Paradox probabilities offer a formal measure of this distortion. Without such contextualization, modern performances may be systematically undervalued. In particular, the modern structure of MLB makes it increasingly unlikely that even players with exceptional latent ability will produce historically dominant observed outcomes.

Supplemental Results

Lemma 5.1. *Let $1 \leq j < n$ and let $x \in (0,1)$. Let $I_{\alpha,\beta}(x)$ be a regularized incomplete beta function as in Definition 3.3. Then, $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$.*

Need to fix notation; should be $I_{\alpha,\beta}(x)$

Proof. We can define a function $f_{n,j}(x)$ as

$$\begin{aligned} f_{n,j}(x) &= I_{n-j,j+1}(x^{\frac{1}{n}}) - I_{n-j+1,j+1}(x^{\frac{1}{n+1}}) \\ &= \int_0^{x^{\frac{1}{n}}} \frac{y^{n-j-1}(1-y)^j}{B(n-j,j+1)}dy - \int_0^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1,j+1)}dy. \end{aligned}$$

An application of the chain rule yields

$$\begin{aligned} f'_{n,j}(x) &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n+1)B(n-j+1,j+1)} \\ &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n-j)B(n-j,j+1)} \\ &= \frac{1}{B(n-j,j+1)}\left(\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j}\right), \end{aligned}$$

where we used $B(n-j+1,j+1) = \frac{n-j}{n+1}B(n-j,j+1)$ in the second equality. Now define $A_{n,j} = \{x : f'_{n,j}(x) \geq 0\}$. For $x \in A_{n,j}$ we have

$$\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} \geq \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j}$$

$$\implies \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^{j}\left(x^{\frac{1-n}{n}}\right)}{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j}\left(x^{-\frac{n}{n+1}}\right)} \geq \frac{n}{n-j}$$

$$\implies x^{-\frac{j}{n(n+1)}}\left(\frac{1-x^{\frac{1}{n}}}{1-x^{\frac{1}{n+1}}}\right)^{j} \geq \frac{n}{n-j}$$

$$\implies \frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}} \geq \left(\frac{n}{n-j}\right)^{\frac{1}{j}}.$$

Define $g_n(x) = \frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}}$. Then

$$g_n(x) = \frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}} = \frac{x^{-\frac{1}{n(n+1)}}-1}{1-x^{\frac{1}{n+1}}}+1,$$

and

$$\begin{aligned} g_n'(x) &= \frac{1}{(1-x^{\frac{1}{n+1}})^2}\left(-\frac{1}{n(n+1)}x^{-\frac{1}{n(n+1)}-1}(1-x^{\frac{1}{n+1}}) - (x^{-\frac{1}{n(n+1)}}-1)(-\frac{1}{n+1}x^{\frac{1}{n+1}-1})\right) \\ &= \frac{1}{(1-x^{\frac{1}{n+1}})^2}\left(\frac{1}{n}x^{-\frac{n^2+1}{n(n+1)}} - \frac{1}{n+1}x^{-\frac{n}{n+1}} - \frac{1}{n(n+1)}x^{-\frac{n^2+n+1}{n(n+1)}}\right) \\ &= \frac{x^{-\frac{n^2+1}{n(n+1)}}}{n(n+1)(1-x^{\frac{1}{n+1}})^2}\left((n+1)-nx^{\frac{1}{n(n+1)}}-x^{-\frac{1}{n+1}}\right). \end{aligned}$$

Now define $l_n(x) = (n+1) - nx^{\frac{1}{n(n+1)}} - x^{-\frac{1}{n+1}}$ and see that

$$l_n'(x) = -\frac{1}{n+1}x^{\frac{1}{n(n+1)}-1} + \frac{1}{n+1}x^{-\frac{1}{n+1}-1} = \frac{1}{n+1}x^{-\frac{n+2}{n+1}}\left(1-x^{\frac{1}{n}}\right).$$

Thus, $l_n(x)$ is an increasing function for $x \in (0,1)$ and $\sup_{x\in(0,1)} l_n(x) = 0$. Therefore, $l_n(x) < 0$ for $x \in (0,1)$. Thus, $g_n(x)$ is an decreasing function, and $g_n(x) \to 1$ as $x \to 1$. We can see that $g_n(x) \in (1,+\infty)$ for $x \in (0,1)$. Thus $A_{n,j} = (0, x^*_{n,j}]$ where $x^*_{n,j}$ is such that $g_n(x^*_{n,j}) = (\frac{n}{n-j})^j$. Putting it all together, $f_{n,j}(0) = f_{n,j}(1) = 0$ and this implies that $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$ for all $x \in (0,1)$. □

Proposition 5.2. *Consider a Full House Model in Definition 2.1 with two populations and $F_{Y_i} = F_Y$ for $i = 1,2$ where the system inclusion function is $1_i(X_{i,j} \geq X_{i,(N_i-n_i+1)})$. Let I be a regularized incomplete beta function. In the notation of Theorem 3.4, suppose that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $N_2 - N_1 > n_2 - n_1 \geq 0$, and $k = j$. Then, technical condition (4) holds and the conclusion of Theorem 3.4 follows.*

Proof. The conditions of this proposition allow us to express technical condition (4) as

$$f_{n_1,n_2,N_1,N_2,j}(x) = I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x).$$

Applications of the chain rule, the inverse function theorem [Rudin et al., 1964] on Page 221, and this property of regularized incomplete beta functions

$$I'_{\alpha,\beta}(I^{-1}_{\alpha,\beta}(x)) = \frac{I^{-1}_{\alpha,\beta}(x)^{\alpha-1}(1-I^{-1}_{\alpha,\beta}(x))^{\beta-1}}{B(\alpha,\beta)},$$

allow us to write

$$\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(x) &= \frac{I'_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)}{I'_{n_2-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)} - \frac{I'_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)}{I'_{N_2-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)} \\
&= \frac{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_1-j,j+1)}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_2-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_2-j,j+1)} \\
&\quad - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_1-j,j+1)}{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_2-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_2-j,j+1)} \\
&= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)}\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)}\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2} \\
&= \left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}\frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&\quad \times \left[\frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}\right]
\end{aligned}$$

From here we see that $f_{n_1,n_2,N_1,N_2,j}(0) = f_{n_1,n_2,N_1,N_2,j}(1) = 0$, and when $x = 1$, we have

$$\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(1) &= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&= \frac{(n_1-j)\ldots(n_1)}{(n_2-j)\ldots(n_2)} - \frac{(N_1-j)\ldots(N_1)}{(N_2-j)\ldots(N_2)} \\
&\leq 0,
\end{aligned}$$

from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$. If we can prove the $\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$ is a monotonic decreasing function for $x \in (0,1)$ and ranges from 0 to 1, then we can conclude that $f_{n_1,n_2,N_1,N_2,j}(x) \geq 0$ as desired.

Define $r(x) = r_{n_1,n_2,N_1,N_2,j}(x) = \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$, and notice that

$$r_{n_1,n_2,N_1,N_2,j}(x) = \left(\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{a}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)}\right)^{b},$$

where $a = \frac{N_1-N_2}{n_1-n_2}$ and $b = n_1 - n_2$. The conditions of the proposition imply that $b \leq 0$ and $1 \leq a \leq \frac{N_2}{n_2}$. When $a = 1$ we have that $r(x) \in (0,1)$ since $I^{-1}_{N_2-j,j+1}(x) > I^{-1}_{n_2-j,j+1}(x)$.

Now let $s_a(x) = s_{n_2,N_2,a}(x) = \frac{I^{-1}_{N_2-j,j+1}(x)^a}{I^{-1}_{n_2-j,j+1}(x)}$. Then, when $a = \frac{N_2}{n_2}$, we have that

$$s_{\frac{N_2}{n_2}}(x) = \frac{I^{-1}_{N_2-j,j+1}(x)^{\frac{N_2}{n_2}}}{I^{-1}_{n_2-j,j+1}(x)} = \left(\frac{I^{-1}_{N_2-j,j+1}(x)^{N_2}}{I^{-1}_{n_2-j,j+1}(x)^{n_2}}\right)^{\frac{1}{n_2}}.$$

Lemma 5.1 states that for all $x \in (0,1)$, $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$, and this implies that $I^{-1}_{N_2-j,j+1}(x)^{N_2} \geq I^{-1}_{n_2-j,j+1}(x)^{n_2}$. Thus, $s_{\frac{N_2}{n_2}}(x) \geq 1$ for all $x \in (0,1)$.

Need to make coherent

Since $I^{-1}_{x^{\frac{1}{n}}}(n-j,j+1) \in (0,1)$, then for any $a \in (1, \frac{N_2}{n_2})$, $s_a(x) \geq 1$, and this implies that $r(x) \in (0,1)$.

Then we show $\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$ is a monotonic decreasing function for $x \in (0,1)$

Need to make coherent

We can write

$$\begin{aligned}
s_a'(x) &= \frac{a(I^{-1}_{N_2-j,j+1}(x))^{a-1} I^{-1}_{n_2-j,j+1}(x)/I'_{N_2-j,j+1}(I^{-1}_{N_2-j,j+1}(x)) - (I^{-1}_{N_2-j,j+1}(x))^a/I'_{n_2-j,j+1}(I^{-1}_{n_2-j,j+1}(x))}{I^{-1}_{n_2-j,j+1}(x)^2} \\
&= T_a(x)\left(aI^{-1}_{n_2-j,j+1}(x)I'_{n_2-j,j+1}(I^{-1}_{n_2-j,j+1}(x)) - I^{-1}_{N_2-j,j+1}(x)I'_{N_2-j,j+1}(I^{-1}_{N_2-j,j+1}(x))\right) \\
&= T_a(x)\left(a\frac{I^{-1}_{n_2-j,j+1}(x)^{n_2-j}(1-I^{-1}_{n_2-j,j+1}(x))^j}{B(n_2-j,j+1)} - \frac{I^{-1}_{N_2-j,j+1}(x)^{N_2-j}(1-I^{-1}_{N_2-j,j+1}(x))^j}{B(N_2-j,j+1)}\right),
\end{aligned}$$

where

$$T_a(x) = T_{a,n_2,N_2,j}(x) = \frac{I^{-1}_{N_2-j,j+1}(x)^{a-1}}{I^{-1}_{n_2-j,j+1}(x)^2 I'_{N_2-j,j+1}(I^{-1}_{N_2-j,j+1}(x)) I'_{n_2-j,j+1}(I^{-1}_{n_2-j,j+1}(x))}.$$

Then,

$$s'_{\frac{N_2}{n_2}}(x) = N_2 T_{\frac{N_2}{n_2}}(x)\left(\frac{I^{-1}_{n_2-j,j+1}(x)^{n_2-j}(1-I^{-1}_{n_2-j,j+1}(x))^j}{n_2 B(n_2-j,j+1)} - \frac{I^{-1}_{N_2-j,j+1}(x)^{N_2-j}(1-I^{-1}_{N_2-j,j+1}(x))^j}{N_2 B(N_2-j,j+1)}\right).$$

Now let

$$m(x) = m_{n_2,N_2,j}(x) = \frac{I^{-1}_{n_2-j,j+1}(x)^{n_2-j}(1-I^{-1}_{n_2-j,j+1}(x))^j}{n_2 B(n_2-j,j+1)} - \frac{I^{-1}_{N_2-j,j+1}(x)^{N_2-j}(1-I^{-1}_{N_2-j,j+1}(x))^j}{N_2 B(N_2-j,j+1)}.$$

We can write

$$
\begin{aligned}
m'(x) = \frac{1}{n_2}\Bigg(&\frac{(n_2-j)I^{-1}_{n_2-j,j+1}(x)^{n_2-j-1}}{I^{-1}_{n_2-j,j+1}(x)^{n_2-j-1}(1-I^{-1}_{n_2-j,j+1}(x))^j}(1-I^{-1}_{n_2-j,j+1}(x))^j \\
&- \frac{jI^{-1}_{n_2-j,j+1}(x)^{n_2-j}}{I^{-1}_{n_2-j,j+1}(x)^{n_2-j-1}(1-I^{-1}_{n_2-j,j+1}(x))^j}(1-I^{-1}_{n_2-j,j+1}(x))^{j-1}\Bigg) \\
-\frac{1}{N_2}\Bigg(&\frac{(N_2-j)I^{-1}_{N_2-j,j+1}(x)^{N_2-j-1}}{I^{-1}_{N_2-j,j+1}(x)^{N_2-j-1}(1-I^{-1}_{N_2-j,j+1}(x))^j}(1-I^{-1}_{N_2-j,j+1}(x))^j \\
&- \frac{jI^{-1}_{N_2-j,j+1}(x)^{N_2-j}}{I^{-1}_{N_2-j,j+1}(x)^{N_2-j-1}(1-I^{-1}_{N_2-j,j+1}(x))^j}(1-I^{-1}_{N_2-j,j+1}(x))^{j-1}\Bigg) \\
= \frac{1}{n_2}&\left((n_2-j)-\frac{jI^{-1}_{n_2-j,j+1}(x)}{1-I^{-1}_{n_2-j,j+1}(x)}\right)-\frac{1}{N_2}\left((N_2-j)-\frac{jI^{-1}_{N_2-j,j+1}(x)}{1-I^{-1}_{N_2-j,j+1}(x)}\right) \\
= j&\left(\frac{1}{N_2(1-I^{-1}_{N_2-j,j+1}(x))}-\frac{1}{n_2(1-I^{-1}_{n_2-j,j+1}(x))}\right).
\end{aligned}
$$

We have that $m(0) = m(1) = 0$, and $m'(0) < 0$.

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Also $\lim_{x\to 1} m'(x) > 0$ as $\frac{1}{1-I^{-1}_{N_2-j,j+1}(x)} > \frac{1}{1-I^{-1}_{n_2-j,j+1}(x)}$. In details,

for $\forall\epsilon_1 > 0$ and $\forall\epsilon_2 > 0$, we can find $x \in (0,1)$ that satisfies $I^{-1}_{N_2-j,j+1}(x) > I^{-1}_{n_2-j,j+1}(x)-\epsilon_1$ and $1-x > \epsilon_2$. We choose the $\epsilon = min(\epsilon_1, \epsilon_2)$

Then we can find $x > I^{-1}_{N_2-j,j+1}(1+\frac{\epsilon}{(n_2-N_2)/n_2})$,

$$
\begin{aligned}
\frac{1}{N_2(1-I^{-1}_{N_2-j,j+1}(x))}-\frac{1}{n_2(1-I^{-1}_{n_2-j,j+1}(x))} &= \frac{n_2(1-I^{-1}_{n_2-j,j+1}(x))-N_2(1-I^{-1}_{N_2-j,j+1}(x))}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))} \\
&> \frac{(n_2-N_2)(1-I^{-1}_{N_2-j,j+1}(x))+n_2\epsilon}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))} \\
&> \frac{(n_2-N_2)(-\frac{\epsilon}{(n_2-N_2)/n_2})+n_2\epsilon}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))} \\
&> 0
\end{aligned}
$$

Therefore, $\lim_{x\to 1} m'(x) > 0$.

Define

$$
w_{n_2,N_2}(x) = \frac{1}{N_2(1-I^{-1}_{N_2-j,j+1}(x))}-\frac{1}{n_2(1-I^{-1}_{n_2-j,j+1}(x))} = \frac{n_2w_1x-N_2w_2x}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))}
$$

where $w_1(x) = 1-I^{-1}_{n_2-j,j+1}(x)$, $w_2(x) = 1-I^{-1}_{N_2-j,j+1}(x)$

Since $w_1(x) > w_2(x)$ and $w_2(x)$ decays slower than for $w_2(x)$ for $x \in (0,1)$. Also, we know $w_{n_2,N_2}(0) < 0$.

Thus, $m(x) < 0$ for $x \in (0,1)$.

Now we show $r(x)$ is a decreasing function that ranges from 0 to 1.

Let $x^* = x^*_{n_1,n_2,N_1,N_2,j}$ be such that $f'_{n_1,n_2,N_1,N_2,j}(x^*) = 0$, then

$$r(x^*) = \frac{\left(I^{-1}_{N_2-j,j+1}(x^*)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x^*)\right)^{n_1-n_2}} = \frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} = c,$$

where $c = c_{n_1,n_2,N_1,N_2,j} \in (0,1)$, again from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$.

Then, based on the property of $r(x)$, x^* is the unique value that satisfies the $r(x^*) = c$.

We can conclude $f(x) \geq 0$ for all $x \in (0,1)$.

Is this needed: Let $x^* = x^*_{n_1,n_2,N_1,N_2,j}$ be such that $f'_{n_1,n_2,N_1,N_2,j}(x^*) = 0$, then

$$\frac{\left(I^{-1}_{N_2-j,j+1}(x^*)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x^*)\right)^{n_1-n_2}} = \frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} = c,$$

where $c = c_{n_1,n_2,N_1,N_2,j} \in (0,1)$, again from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$.

□

Proposition 5.3. *Suppose that conditions of Theorem 3.4 where the system inclusion function is $1_i(X_{i,j} \geq X_{(N_i-n_i+1)})$, $i = 1,2$. Suppose that $n_2 \geq n_1 > 1$, $N_1 \geq N_2 > 1$, and fix $k = j$. Then condition (4) holds, and the conclusions of Theorem 3.4 follow.*

Proof. For ease of notation we will let $g_{n,i}(x) = I_{i,n-i+1}(x)$. The proof will use two induction arguments. For the base case let $n_1 = n_2$ and $N_1 = N_2$. With these specifications we have that

$$g_{n_1,n_1-j} \circ g^{-1}_{n_2,n_2-j}(x) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(x) = 0.$$

For the induction step, choose some $m \leq n_1 - 1$ and suppose that

$$g_{n_1-m,n_1-m-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \geq 0, \tag{14}$$

for $t \in (0,1)$ where $n_2 \geq n_1$ and $N_1 \geq N_2$. An integration by parts calculation applied to regularized incomplete beta functions gives

$$I_{\alpha,\beta}(x) = I_{\alpha+1,\beta}(x) + \frac{x^\alpha(1-x)^\beta}{\alpha B(\alpha,\beta)}.$$

Applying the above to (14) yields

$$\begin{aligned} & g_{n_1-m-1,n_1-m-1-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \\ &= g_{n_1-m,n_1-m-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \\ &\quad + \frac{\left[g^{-1}_{n_2,n_2-j}(t)\right]^{n_1-m-1-j}\left[1-g^{-1}_{n_2,n_2-j}(t)\right]^{j+1}}{(n_1-m-1-j)B(n_1-m-1-j,j+1)} \\ &\geq 0. \end{aligned}$$

A similar argument holds for iterative increases to N_1. Thus, condition (4) holds under are assumed conditions and the conclusions of Theorem 3.4 follow. □

When $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$ and $k = j$, then

$g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0,1)$, where $g_{n,i}(x) = I_x(i, n-i+1)$, $I_x(\alpha,\beta)$ is the CDF of Beta(α,β) distribution.

6 Proof to Lemma 4.4

Proof. Notice that the cumulative distribution function of multivariate normal distribution can be decomposed into a series of product of conditional univariate normal distribution.

$$\begin{aligned} \mathbf{\Phi}(\mathbf{W},\mathbf{\Sigma}) = \Phi(W_n = w_n | W_1 = w_1, W_2 = w_2, \ldots, W_{n-1} = w_{n-1}) \times \\ \Phi(W_{n-1} = w_{n-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n-2} = w_{n-2}) \times \cdots \times \\ \Phi(W_2 = w_2 | W_1 = w_1) \times \Phi(W_1 = w_1), \end{aligned} \tag{15}$$

where $\mathbf{W} = (W_1, W_2, \ldots, W_n)$, $\mathbf{w} = (w_1, w_2, \ldots, w_n)$.

Then in general, for any $k \in \{2, \ldots, n_1 + n_2 - 1\}$, the conditional distribution of W_k given $W_1, \ldots, W_{k-1}$ is

$$W_k \mid W_1 = w_1, \ldots, W_{k-1} = w_{k-1} \sim \mathcal{N}\left(\mu_k, \sigma_k^2\right)$$

with

$$\mu_k = \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)}$$

$$\sigma_k^2 = \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{\Sigma}_{1:(k-1),k}$$

Here: $\mathbf{\Sigma}_{k,1:(k-1)}$ is a $1 \times (k-1)$ row vector: the covariances between W_k and $W_1, \ldots, W_{k-1}$ $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ is a $(k-1) \times (k-1)$ covariance matrix, $\mathbf{\Sigma}_{k,k}$ is the kth element on the diagonal of the covariance matrix $\mathbf{\Sigma}$, and $\mathbf{w}_{1:(k-1)} = (w_1, \ldots, w_{k-1})^T$.

Then we will show the the parameters μ and σ^2 of the conditional univariate normal distribution in the set of Equation (11) are finite.

When $k \leq n_1$, by applying the Equation (10)

$\mu_k = \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)}$

$$
\begin{aligned}
&= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}^T \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{(k-1)\times(k-1)}^{-1} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}^T \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}(k-2)+1 & -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}(k-2)+1 \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}^T \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_k^2 &= \boldsymbol{\Sigma}_{k,k} - \boldsymbol{\Sigma}_{k,1:(k-1)} \boldsymbol{\Sigma}_{1:(k-1),1:(k-1)}^{-1} \boldsymbol{\Sigma}_{1:(k-1),k} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}^T \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{(k-1)\times(k-1)}^{-1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}^T \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

Then

$$\Phi(W_k = w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1})$$

$$
\begin{aligned}
&= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{\frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}}\right) \\
&= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1}(1-\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}{\sqrt{1+\frac{\sigma_2^2}{\sigma_1^2}(1-\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}}\right) \\
&= \Phi\left(\frac{\frac{\frac{\mu_2-\mu_1}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{1+\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}}\right) \\
&= \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}}\right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}}\right) = \Phi(0) = 0.5
$$

where $\Phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, no matter what the value of k is, $c_k < 1$.

When $k = n_1 + 1$, the only difference from the case $k \le n_1$ is that $\mathbf{\Sigma}_{k,\,1:(k-1)}$ is different. We can then apply a similar method as in the case $k \le n_1$ to compute μ_k and σ_k.

$$
\begin{aligned}
\mu_k &= \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix}\frac{\sigma_2}{\sigma_1}\\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix}^T_{(k-1)\times 1}
\begin{bmatrix}\frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1\end{bmatrix}^{-1}_{(k-1)\times(k-1)}
\begin{bmatrix}\frac{\mu_2-\mu_1}{\sigma_1}\\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1}\end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_k^2 &= \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{\Sigma}_{1:(k-1),k} \\
&= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

Then

$$
\begin{aligned}
&\Phi(W_k = w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) = \Phi\left(- \frac{\mu_2-\mu_1}{\sigma_2} \right)
$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, no matter what the value of k is, $c_k < 1$.

When $k > n_1 + 1$, we first compute $\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}$. Based on the structure of $\mathbf{\Sigma}$ in Equation (6), the $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ can be decomposed into

$$
\mathbf{\Sigma}_{1:(k-1),1:(k-1)} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \tag{16}
$$

defined with the following structure

$$D_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1 \times n_1} \quad D_{12}^T = D_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-n_1-1)\times n_1} \tag{17}$$

$$D_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(k-n_1-1)\times(k-n_1-1)}.$$

By applying the Equation (10) and Schur complement [Zhang, 2006],

$$\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}, \tag{18}$$

$$E_{11} = (D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \cdot \frac{1}{k-n_1} \cdot \begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix} \right.$$

$$\left. \cdot \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1} =$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \dots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \dots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 \end{bmatrix}^{-1}$$

$$= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}_{n_1\times n_1}$$

$$E_{12} = -(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$
$$\times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 2 \end{bmatrix}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$
$$\times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{k-n_1} \begin{bmatrix} k-n_1-1 & -1 & \cdots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & k-n_1-1 \end{bmatrix}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$
$$\times \begin{bmatrix} \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{n_1\times(k-n_1-1)}$$

Since $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ is a symmetric matrix, then $E_{12} = E_{21}^T$. Thus,

$$E_{21} = -D_{22}^{-1}D_{21}(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$
$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1-1)\times n_1}$$

$$E_{22} = D_{22}^{-1} + D_{22}^{-1}D_{21}(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1}$$
$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$
$$= \frac{1}{k-n_1}\begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix}$$
$$+ \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \end{bmatrix}$$

$$= \frac{1}{k-n_1}\begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}$$

$$= \begin{bmatrix} 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1)\times(k-n_1)}$$

Then

$$\mu_k = \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(k-1)\times 1}$$

$$= \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22}\right] \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1(k-n_1-1)}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$\sigma_k^2 = \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{\Sigma}_{1:(k-1),k}$$

$$= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{(k-1)\times 1}$$

$$= 2 - \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right] \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$= 2 - \left(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right.$$

$$
\begin{aligned}
&+\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix} E_{12} \begin{bmatrix}1\\ \vdots \\ 1\end{bmatrix} + \begin{bmatrix}1 & \dots & 1\end{bmatrix} E_{22} \begin{bmatrix}1\\ \vdots \\ 1\end{bmatrix}\Bigg) \\
&= 2 - \Bigg(\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \begin{bmatrix}\frac{\sigma_2}{\sigma_1}\\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix} \\
&+\begin{bmatrix}1 & \dots & 1\end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}\frac{\sigma_2}{\sigma_1}\\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix} \\
&+\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}1\\ \vdots \\ 1\end{bmatrix} \\
&+\begin{bmatrix}1 & \dots & 1\end{bmatrix} \begin{bmatrix} 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}1\\ \vdots \\ 1\end{bmatrix}\Bigg) \\
&= 2 - \Bigg(\begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}\frac{\sigma_2}{\sigma_1}\\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}\frac{\sigma_2}{\sigma_1}\\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix} \\
&- \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}1\\ \vdots \\ 1\end{bmatrix} + \begin{bmatrix} 1-\frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & 1-\frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix}1\\ \vdots \\ 1\end{bmatrix}\Bigg) \\
&= 2 - \Big(\frac{\frac{\sigma_2^2}{\sigma_1^2}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{2(k-n_1-1)n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} + \frac{k-n_1-1}{k-n_1} + \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{(k-n_1-1)^2 n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \Big)
\end{aligned}
$$

$$= 2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

Then

$$\begin{aligned}
&\Phi(W_k = w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}} \right) \\
&= \Phi\left(- \frac{\frac{\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}} \right) \\
&= \Phi\left(- \frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1)+\sigma_1^2(k-n_1)^2}}} \right) \\
&= c_k
\end{aligned}$$

$$\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(- \frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1)+\sigma_1^2(k-n_1)^2}}} \right) = \Phi(0) = 0.5$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, no matter what the value of k is, $c_k < 1$.

Therefore, we show the parameters μ and σ^2 of the conditional univariate normal distribution in the set of Equation (11) are finite.

Let

$$c_{max} = \max_{k=2,\ldots,n_1+n_2-1} c_k,$$

then $c_{max} < 1$.

We can rewrite $\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma})$

$$\begin{aligned}
\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) &= \Phi(W_n = w_n | W_1 = w_1, W_2 = w_2, \ldots, W_{n-1} = w_{n-1}) \times \\
&\Phi(W_{n-1} = w_{n-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n-2} = w_{n-2}) \times \cdots \times \\
&\Phi(W_2 = w_2 | W_1 = w_1) \times \Phi(W_1 = w_1) \\
&= \Phi(W_1 = w_1) \times \Pi_{k=2}^{n_1+n_2-1} c_k
\end{aligned}$$

Then the set

$$\begin{aligned}
&\Big\{\Phi(W_{n_1+n_2-1} = w_{n_1+n_2-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-2} = w_{n_1+n_2-2}), \\
&\Phi(W_{n_1+n_2-2} = w_{n_1+n_2-2} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-3} = w_{n_1+n_2-3}), \ldots, \\
&\Phi(W_2 = w_2 | W_1 = w_1), \Phi(W_1 = w_1)\Big\} \\
&= \Big\{\Phi(W_1 = w_1), c_2, \ldots, c_{n_1+n_2-1}\Big\}
\end{aligned} \tag{19}$$

,

Therefore,

$$\begin{aligned}
&\max\Big\{\Phi(W_{n_1+n_2-1} = w_{n_1+n_2-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-2} = w_{n_1+n_2-2}), \\
&\Phi(W_{n_1+n_2-2} = w_{n_1+n_2-2} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-3} = w_{n_1+n_2-3}), \ldots, \\
&\Phi(W_2 = w_2 | W_1 = w_1), \Phi(W_1 = w_1)\Big\} \\
&= \max\Big\{\Phi(W_1 = w_1), c_2, \ldots, c_{n_1+n_2-1}\Big\} \\
&= \max\Big\{\Phi(W_1 = w_1), c_{max}\Big\} \\
&:= c
\end{aligned} \tag{20}$$

Since all terms in the set above are the cumulative distribution functions of univariate normal distribution and $\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)$, then c is strictly less than 1.

$$\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) \leq c^{n_1+n_2-1} \tag{21}$$

□

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Year	Name	MLB leading BA	n	Shapiro Wilk p-value	Extreme Percentile
1876	Ross Barnes	0.429	40	0.0043	0.9791
1881	Cap Anson	0.399	42	0.3213	0.9665
1884	Fred Dunlap	0.412	110	0.0066	0.9722
1887	Tip O'Neill	0.435	112	0.0352	0.9848
1901	Nap Lajoie	0.426	104	0.3360	0.9760
1904	Nap Lajoie	0.376	104	0.0402	0.9861
1905	Cy Seymour	0.377	112	0.0437	0.9602
1908	Honus Wagner	0.354	115	0.0462	0.9614
1909	Ty Cobb	0.377	111	0.0937	0.9673
1910	Nap Lajoie	0.383	118	0.0082	0.9775
1911	Ty Cobb	0.420	115	0.0416	0.9663
1912	Ty Cobb	0.409	126	0.0001	0.9730
1913	Ty Cobb	0.390	119	0.0132	0.9570
1916	Tris Speaker	0.386	114	0.0003	0.9915
1917	Ty Cobb	0.383	117	0.0890	0.9724
1918	Ty Cobb	0.382	109	0.4605	0.9575
1922	George Sisler	0.420	118	0.0174	0.9671
1924	Rogers Hornsby	0.424	118	0.1749	0.9863
1941	Ted Williams	0.406	110	0.0104	0.9946
1957	Ted Williams	0.388	117	0.0534	0.9885
1977	Rod Carew	0.388	196	0.2965	0.9935
1980	George Brett	0.390	184	0.0400	0.9960
1985	Wade Boggs	0.368	190	0.0417	0.9885
1988	Wade Boggs	0.366	180	0.0573	0.9789
1994	Tony Gwynn	0.394	153	0.0392	0.9631
2002	Barry Bonds	0.370	209	0.0102	0.9647
2004	Ichiro Suzuki	0.372	207	0.0622	0.9634
2023	Luis Arraez	0.354	205	0.0055	0.9564

Table 1: Batting average performances that are at least as extreme as Luis Arraez in 2023. Extreme as defined by the value $\widehat{F}(Y_{(n_i),i})^{n_i}$ where i denotes the season, n_i denotes the number of full-time players in season i, $Y_{(n_i),i}$ denotes the highest batting average for full-time players in season i, and $\widehat{F}$ is the distribution function of the normal distribution with sample mean and standard deviation estimators substituted for the population parameters. This table also includes the year, name, and raw batting average of extreme performers. It also includes the number of full-time players (n), and a Shapiro Wilk p-value corresponding to a test against normality for the batting average distribution for full-time players.